

Non-commutative ring

Ring of matrices:

$$M(2, \mathbb{Z}) : \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 17 & 9 \\ 19 & -10 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 6 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 9 \\ 4 & 3 \end{pmatrix}$$

$$1 \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad 0 \approx \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad -1 \approx \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

① rational numbers  
+, -, ×, ÷

rational number

with notion  
of equality:

$$\frac{a}{b} = \frac{c}{d} \Leftrightarrow$$

$$\frac{2}{3} = \frac{4}{6}$$